

Blood must be collected with care and adequate safety precautions to ensure:

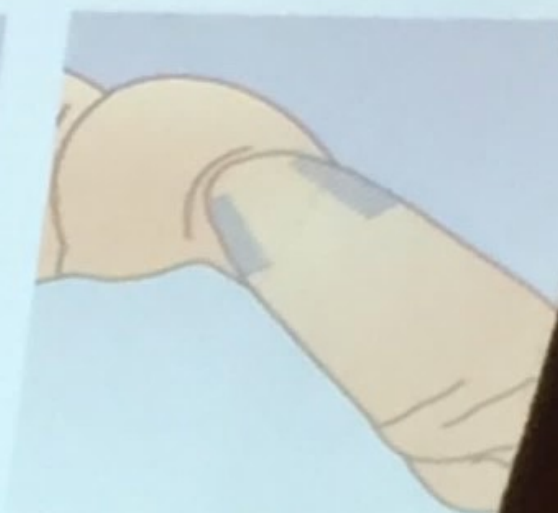
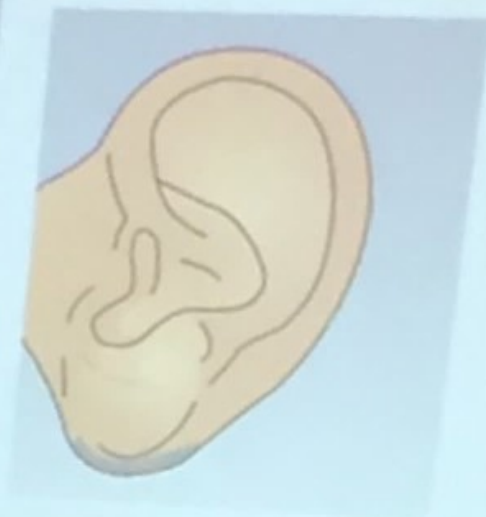
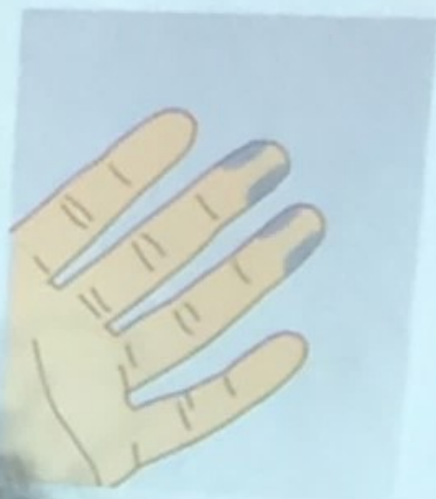
- Test results are reliable
- Contamination of the test sample is avoided and
- Infection from the blood transmissible pathogens is prevented.

Blood can be collected from 2 different *sources*

- Venous blood.
- Capillary blood.

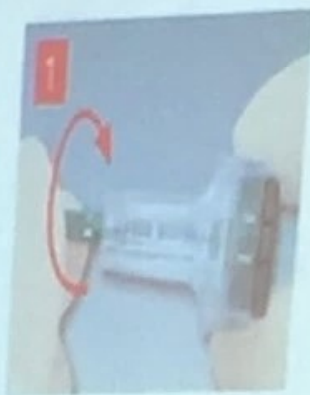
Capillary blood collection.

- A small amount
- In Children and Adults, finger puncture
- in Infants, Heel puncture
- where venipuncture is difficult



REQUIREMENTS FOR THE FINGER PUNCTURE

- Blood Lancet
- Alcohol/Spirit Swab
- Surgical Gauze
- Frosted Slide
- Capillary tube
- Glass Marking pencil



1. Twist the protective cap until it separates from the Safety-Lancet.



2. Press the Safety-Lancet firmly against the chosen and cleaned puncture site and press the firing button.



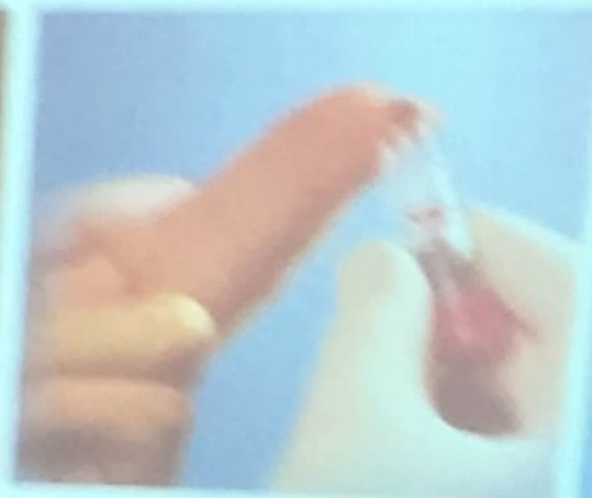
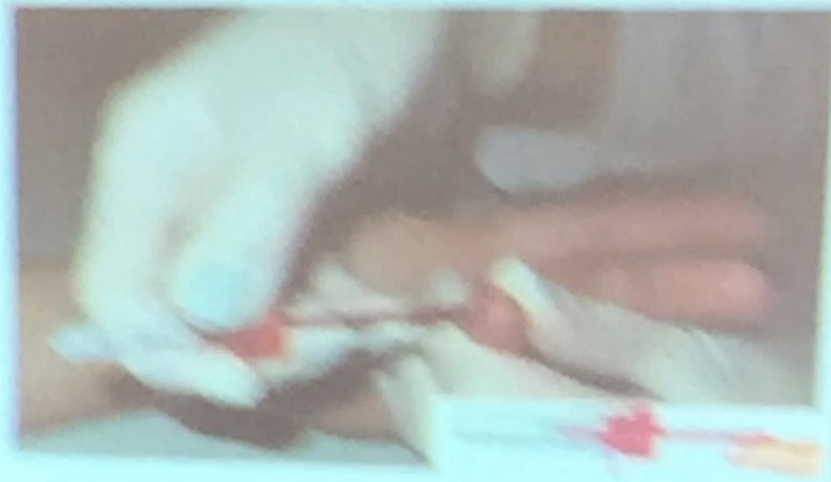
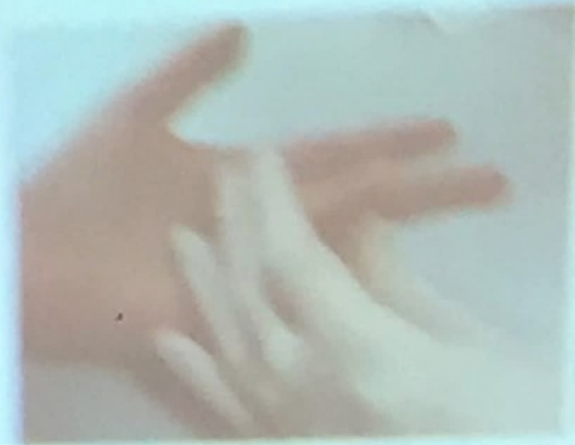
3. Dispose of the Safety-Lancet into a sharps container.

PROCEDURE OF THE FINGER PUNCTURE

- Assembly of the required materials
- Identify the patient and ensure that the patient should be in a comfortable position.
- Find a spot on the Middle or Ring finger of the left hand and warm the site which increase blood flow.
- Disinfect the site of puncture with a sterile cotton swab dipped in 70% Isopropyl Alcohol, to remove the Dust or Epithelial Debris (Dead Cells on Skin).
- Grasp the finger tightly and do the puncture with a sterile lancet. The puncture should be 2-3 mm deep

Cont...

- Wipe out the first drop of blood to avoid tissue fluid contamination & apply gentle pressure to obtain a free flow of blood.
- Blood drops can be taken onto the Clean & grease free glass slide/capillary tube for making the blood smear
- When the desired amount of blood is obtained, stop bleeding by using a clean and dry cotton swab.



Venous blood

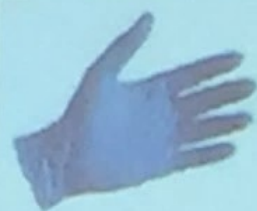
Most commonly required

- ✓ Because most majority of routine tests are performed on venous blood.
- ✓ Blood can be taken directly from the vein.
- ✓ The best site for venous collection is the deep veins of the **antecubital fossa**.
- ✓ Venipuncture can be done by two methods:
 - Syringe method / Open Method
 - Vacutainer method / Closed method

Venipuncture Equipment

- Tourniquet
- Needle/syringe
- Vacutainer holder
- Vacutainer tubes
- Winged infusion set (Butterfly needle)
- Alcohol
- Gauze
- Band aid
- Sharps container
- Marker for tubes

The Equipment Necessary to Perform a Venipuncture



Gloves



Needles



The Hub



Evacuated Collection Tubes



Alcohol Wipes



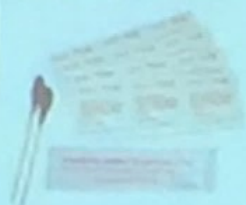
Syringes



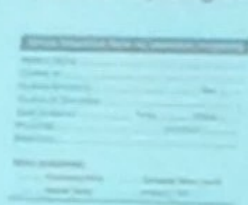
Bandages/Tape



Gauze Sponges



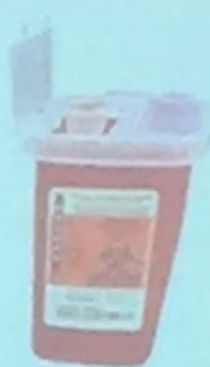
Povidone/Iodine Swabs/Wipes



Requisition Form



Tourniquet



Sharps Container

cephalic vein

brachial artery

(Caution: risk of
unintentional
arterial puncture)

median cubital vein

basilic vein

median vein



Step by Step Procedure

- Put on PPE
- All supplies within easy reach, Assemble needle and holder
- Apply tourniquet
- Select site and cleanse with alcohol
- Remove needle cover
- Pull down skin to anchor vein
- Penetrate skin with bevel of needle up
- Push on tubes, release tourniquet, apply gauze and pressure
- Apply band aid, label tubes

BLOOD SMEAR

- A basic and a highly informative hematological tool
 - In screening,
 - diagnosis and
 - monitoring of disease progression and
 - therapeutic response
- Made from capillary blood or from a drop of blood from an EDTA (anticoagulated) blood sample
- Two types of blood smears
 - Thin
 - Thick

Three basic steps to make blood film

1. Preparation of blood smear.
2. Fixation of blood smear.
3. Staining of blood smear.

Differences Between Thick and Thin Blood Smear

Thick Blood Smear	Thin Blood Smear
Most useful for detecting the presence of parasites	Identify species of parasite causing the infection.
drop of blood on a glass slide(small area)	drop of blood that is spread across a large area of the slide.
WBC, platelets and parasites are visualized	RBC are visible which helps for comparison b/n infected & not infected
allow a more efficient detection of parasites (increased sensitivity 30 X than thin smear).	Less sensitive than a thick film especially where there is a low parasitemia.
It is not fixed in methanol	fixed in methanol
estimate parasitemia	quantify parasitemia, recognize parasite forms like schizonts and gametocytes and Mixed infections

Preparation of Thin Blood Films

- Can be prepared using:
 - Wedge method (Two-slide method)or
 - Cover glasses method
- The most common method is the wedge or push slide smear technique
 - Slides are not easily broken
 - Slides are easier to label
 - When large numbers of films are to be dealt with, slides will be much easier to handle

Materials Needed

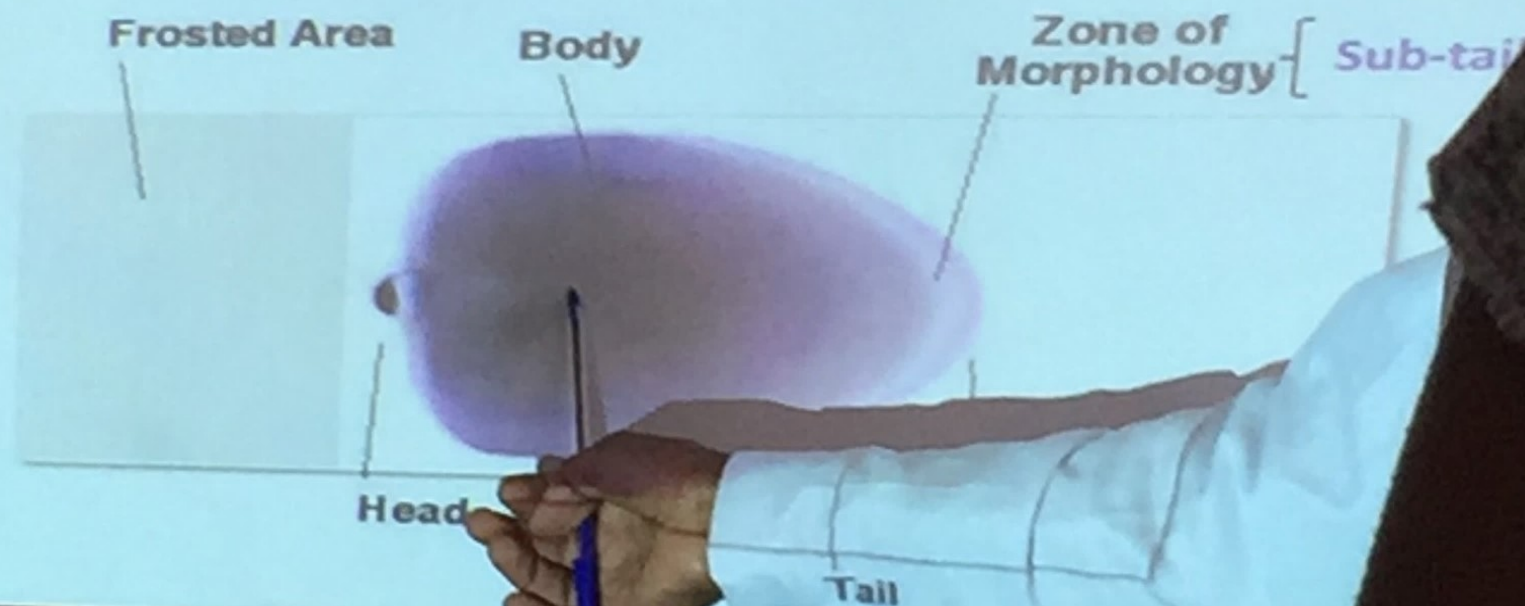
- ◆ PPE
- ◆ Clean glass microscope slides
- ◆ Spreader slide
- ◆ Well-mixed EDTA blood sample/capillary collected blood
- ◆ Device to make a drop of blood
- ◆ Marking pen or pencil
- ◆ Waste and sharps disposal containers

Procedure for preparation of thin blood film

1. Use a glass slide that is free of dust, grease and debris
2. Place a small drop of well mixed EDTA blood 1.0 cm from the end of the glass slide, using either a plain capillary tube or other type of blood dropping device
3. The spreading slide is placed in front of the drop of blood at an angle of about 30° - 40° to the slide and then is moved back to make contact with the drop
4. The drop will spread out quickly along the line of contact of the spreader with the slide

Acceptable Smears

- Smooth, homogeneous and have no lines, vacuoles or holes, jerks, streaks, or ridges
- The length of the film should be $\frac{1}{2}$ to $\frac{3}{4}$ of the length of the slide
- Free of visible clotting
- Not too thick and not too thin
- Gradual transition from the thick to thin areas terminating in smooth 'tail'



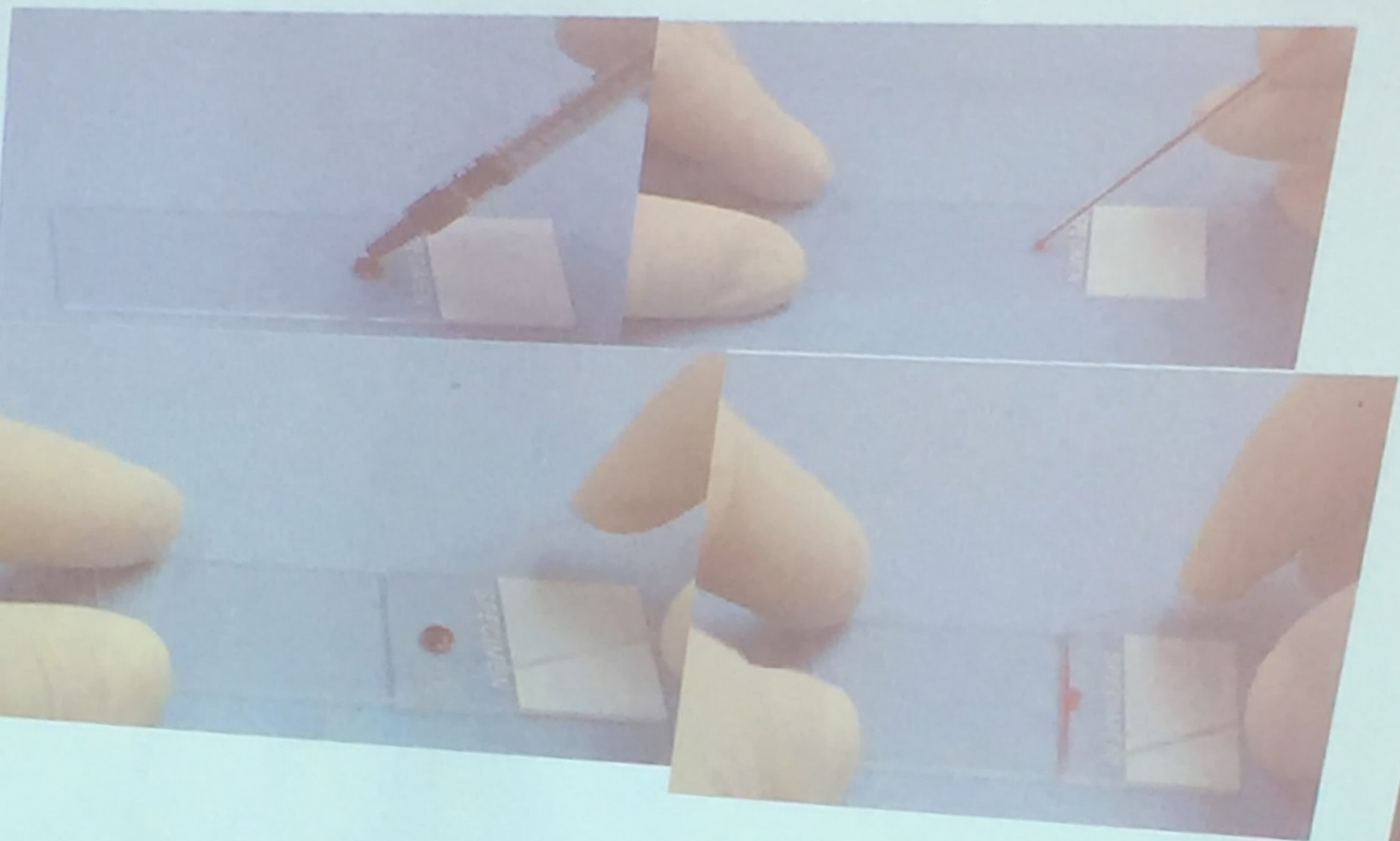
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5. The spreader is advanced with a smooth steady motion so that a thin film of blood is spread over the slide
6. Allow the smear to air-dry
7. Label the name of the patient and date or reference number on the head of the film using a lead pencil

Preparation of thick blood film

1. Place a small drop of blood on a clean slide
 - Could be performed with thin film
2. Spread it with an applicator stick or the corner of another slide until small prints are just visible through the blood smear
3. Allow the smear to air-dry

Steps for blood film preparation



Examples of unacceptable smears



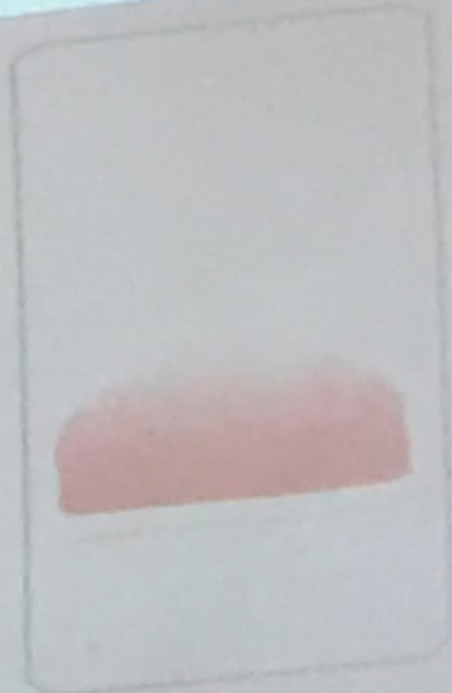
A



B



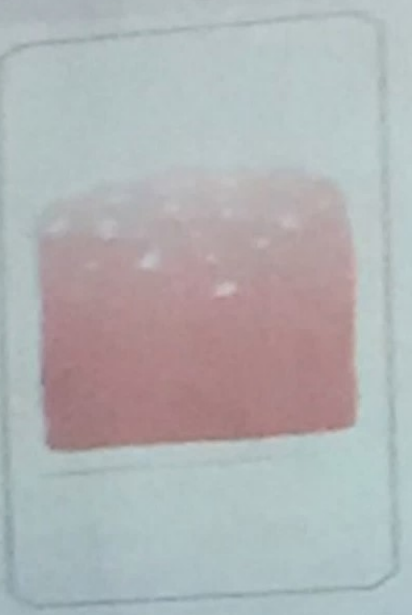
C



D



E



F



G



H

Fixation of blood smear

- To preserve the morphology of the cells, and attach (stick) the cells on the slide
- Methyl alcohol (methanol) is the choice, although ethyl alcohol ("absolute alcohol") can be used.
- Place the film on a staining rack and add 1 – 2 drops of moisture-free methyl alcohol and allow it to dry on the film.
- Alternatively, the blood film can be immersed in a container of absolute methanol for about 2 minutes

Staining of blood smear

- **Most commonly used stains are:**

- **Giemsa**

- Wright's stain

- Leishman's stain

- **Principle of Staining**

- Acidic dyes, basic components(cytoplasm)

- Basic stains, acidic parts(nucleus)

Giemsa Stain

- Giemsa stain is a mixture of Azure, Methylene blue, and Eosin dye.
- Giemsa stain is a differential stain
- The Giemsa stain is one of the **best** stains for malaria and other HP
- Good stain for routine examination of blood smear and used to differentiate nuclear and cytoplasmic morphology of the various cells

PRINCIPLE OF GIEMSA STAINING

- The working principle of the Giemsa stains is
- The acidic dye, Eosin & Azure variably stains the Basic components of the cells i.e. the cytoplasm, Granules etc. and the Basic dye, Methylene blue stains the Acidic components, especially the Nucleus of the cell.

Methods of staining

- Two methods for staining
 - Rapid (10% stain working solution) and
 - Slow (3% stain working solution) methods
- The rapid;
 - commonest method for staining 1–15 slides at a time
 - used in outpatient clinics and busy laboratories where a quick diagnosis is essential
- The slow (3% stain working solution) method
 - larger numbers of slides (≥ 20)
 - cross-sectional or epidemiological surveys and field research
 - less appropriate when a quick result is needed

SUPPLIES AND MATERIALS

- Giemsa stain (3% or 10% solution)
- A small container or beaker for Giemsa working stain;
- Absolute methanol, acetone-free;
- Pasteur pipette with a rubber teat;
- Small container or beaker for methanol;
- Curved plastic staining tray, plate or staining rack;
- Timer;
- slide-drying rack;

Procedure for Giemsa staining

1. Place the slides individually on the staining rack, making sure that they are not touching each other.
2. Pour the stain gently onto the slides until they are totally covered. Each slide will require approximately 3 mL of stain. Avoid pouring the stain directly onto thick films.
3. Leave the stain on the slides for 45-60 min with 3% Giemsa solution and 10-15 min with 10% Giemsa solution.
4. Flood the slides gently with buffered water to float off the iridescent "scum" on the surface of the stain.
5. Remove the slides one by one and place them, thick film downwards, in a drying rack to drain and dry, making sure that the thick film does not touch the edge of the rack.

Cont...

6. Dry it in a vertical position
7. First examined under low power(X10) objective to get an idea of quality of film
8. Then select a suitable area and examine with(X40) and 100X oil immersion

